Machine learning identifies weak prognostic signal in colorectal polyp's transcriptome

Russell Hung[1], Simon Fisher[1], Ditte Andersen[2], Gerard Lynch[3], Noori Maka[4], Jennifer Hay[3], Jakub Jawny[4], William Sloan[4], Stephen McSorley[3], Joanne Edwards[3] and Ian Poole[1]
on behalf of Integrated Technologies for Improved Polyp Surveillance (INCISE) collaborators

Canon
CANON MEDICAL RESEARCH EUROPE LTD.

1. Canon Medical Research Europe, 2. BioClavis,
3. University of Glasgow 4. NHS Greater Glasgow & Clyde

INCISE

Background

Colorectal polyps are growths that can appear on the lining of the colon or rectum and represent possible premalignant lesions to colorectal cancers (CRC). As most polyps are asymptomatic, in the UK surveillance of polyps is monitored through colonoscopies, which is an expensive and unpleasant procedure.

Patients with metachronous polyps may need multiple colonoscopies throughout their lifetime. Understanding the risk factors associated with development of metachronous polyps can result in better risks stratification and management of screening program resources. In this study, we are particularly concerned with identifying distinct patterns in the transcriptome of index polyps related to metachronous risks. We aim to use machine learning and deep learning techniques to:

1. Model occurrence of metachronous polyps based on the index polyp transcriptome
2. Compare binary classification and time-to-event approach in modelling polyp recurrence

This research is part of the Integrated Technologies for Improved Polyp Surveillance[1] (INCISE) project that aims to improve on current bowel cancer screening program in the UK. The INCISE project is a retrospective study that seeks to develop a comprehensive risk stratification tool for metachronous polyps prediction.

Methods

Data Acquisition

The data used in this study is collected as part of the INCISE project. Index polyps are sequenced with TempOSeq to collect gene expression data over three batches. Only patients who underwent a follow-up colonoscopy were selected. The data is composed of expression counts of 22,537 genes for 1,789 patients. Corresponding data on polyp recurrence events and right censored time of event occurrence since index polypectomy for the patients were also obtained.

Data Pre-processing

The data was split into training (1,380) and validation (446) sets. The training set was used in model development and parameter search. The validation set was only used for evaluation of models. Splitting was stratified on event outcome (recurrence vs non-recurrence) and batch identity to account for class imbalance. In the validation set, the minority class within each batch is up-sampled so that there is an even distribution of events in batches. The counts per gene were normalised to counts per million (CPM) and then transformed with $\log_2$ with an offset of 1.

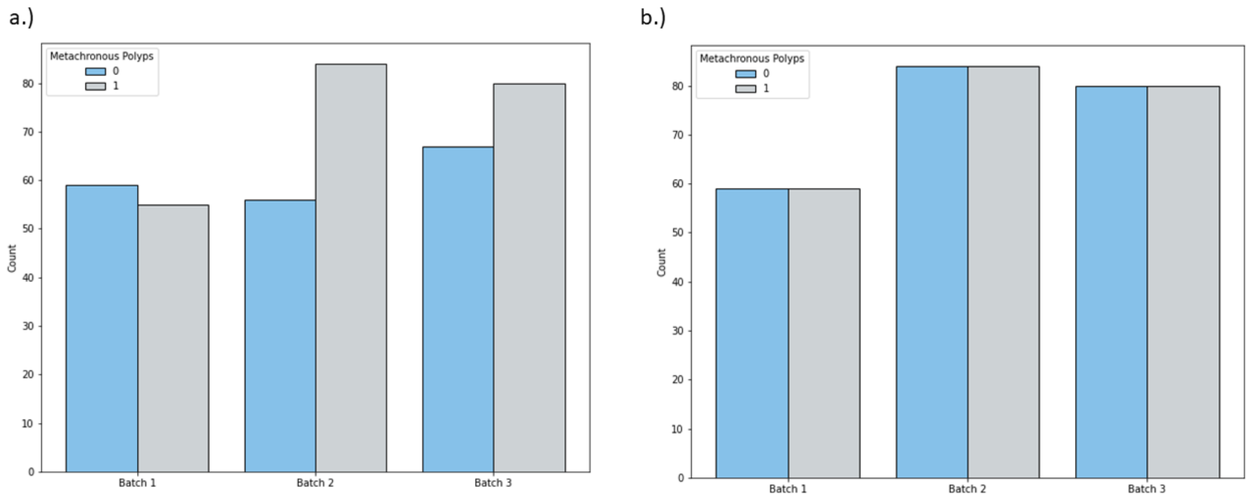

Figure 1. Distribution of positive and negative cases (0 for non-recurrence and 1 for recurrence) in each batch in the holdout validation set before (a) and after (b) up-sampling. Up-sampling ensures that the distribution of labels is even within each batch.

Modelling approach

1. Binary classification approach

We first model the occurrence of metachronous polyps as a binary classification problem. Dimensionality reduction was performed either with principal component analysis (PCA) or by selecting genes with known association with CRC. In the latter case, genes were selected if they are related to the key term 'colorectal cancer' during database search. 3 separate databases were queried – GWAS Catalogue, PGS Catalogue and Protein Atlas.

Logistic regression (LR), support vector machine (SVM) (linear kernel) and random forests (RF) were applied to each dimensionality reduction scheme, giving 12 models. Grid search was used to find the optimal hyperparameters for each model and 5-fold cross validation was used to assess model performance. The model with highest area under the Receiver Operating Characteristics curve (ROC-AUC) was selected to be evaluated on the hold-out validation set.

2. Time-to-event approach

This modelling approach incorporates the time to polyp recurrence. Intuitively, a shorter time to polyp recurrence should be associated with higher risks. An autoencoder with 32 nodes in the bottleneck layer was used to extract latent representations from the gene expression data. Nnet-survival (implemented in PyCox[2] library), a discrete time survival model, was used to model the probabilities of polyp recurrence over a 6-year period from the latent representations.

5-fold cross validation was used to evaluate model performance in the training set. The metric of primary interest is the ROC-AUC at the 5-year mark. The ROC-AUC at each 0.2-year interval from 6 months up to 6 years was also examined. Polyp recurrence event labels were adjusted according to the time under investigation, i.e., a patient having a recurrence 5 years after index polypectomy would be treated as not having recurrence when examining the timepoints before year 5. Optuna[3] was used to perform hyperparameter optimisation. The best model was tested on the holdout validation set.

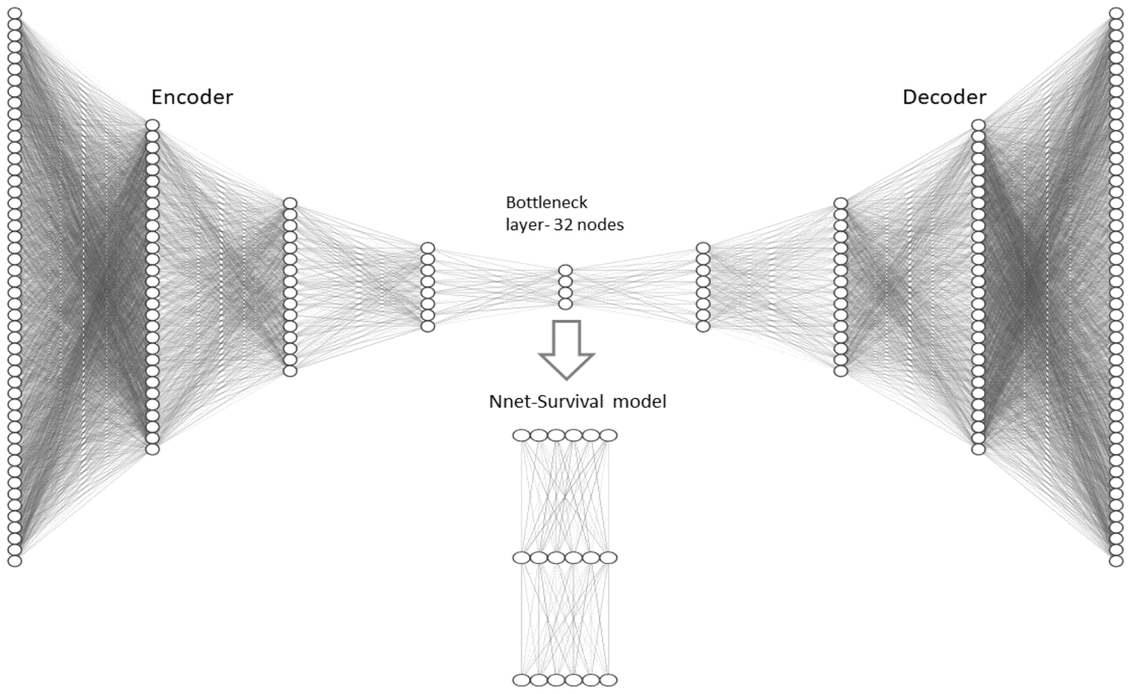

Figure 2. Overall model architecture of autoencoder-survival model. Encoder and decoder are symmetric, and contains 512, 256, 128 and 64 nodes on each respective hidden layer. All layers are 1-dimensional. The bottleneck layer contains 32 nodes. They are used as input features for Nnet-survival model. The model is trained simultaneously on autoencoder reconstruction and time-to-event modelling. Figure produced with NN-SVG[4].

Results

Binary Classification Approach

GWAS-selected genes coupled with SVM and LR provides the best performance in cross-validation study. RF is consistently the worst-performing classifier across all sets of input features. GWAS-SVM model was not predictive in the holdout validation set (ROC-AUC = 0.52).

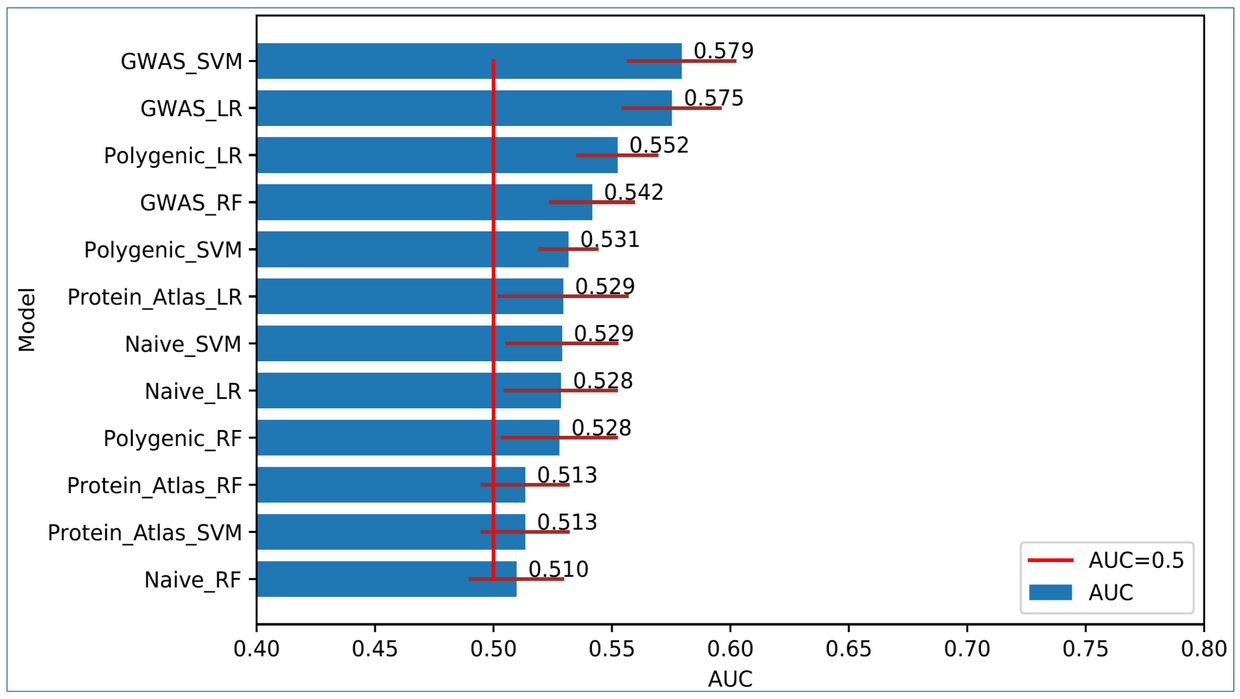

Figure 3. ROC-AUC score with 1 SD error bars for prediction of metachronous polyps given by 12 models, which is derived from combinations of 4 sets of input features on 3 classifies. Abbreviations of model: GWAS Catalogue (GWAS), PGS Catalogue (Polygenic) and Protein Atlas (Protein Atlas), or all genes on PCA (naïve). Support vector machine (SVM), logistic regression (LR), random forests (RF).

Time-to-event approach

The single best model obtained from the cross-validation study had predictive performance of about 0.62 ROC-AUC between 1-year to 6-year from the date of index polypectomy. Model performance differ significantly between re-substitution and holdout validation (approximately 0.90 and 0.54 at 5-year timepoint respectively).

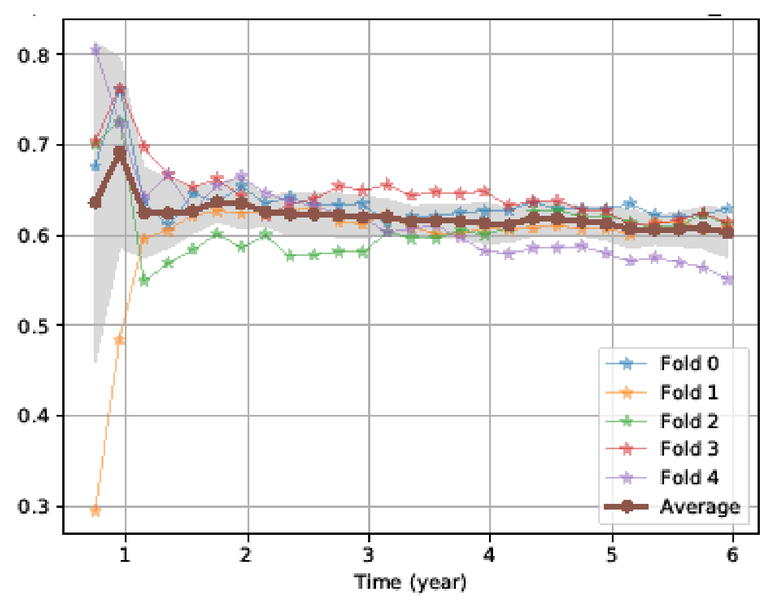

Figure 4. ROC-AUC for future polyp recurrence predicted by time-to-event model over a 6-year period from the date of index polypectomy

Approach	Model	5-fold cross validation AUC	Re-substitution AUC	Holdout Validation AUC
Binary Classification	GWAS-SVM	0.58	0.70	0.52
Time-to-event	Autoencoder-Nnet survival	0.62	0.90	0.54

Table 1. Comparison of model performance between binary classification and time-to-event approach. Scores reported are ROC-AUC. For time-to-event approach, the ROC-AUC at the 5-year timepoint is reported.

Key Findings and Conclusions

Overall, weak signals from both models in the holdout validation set indicates that neither approach was successful in prognostic prediction. The large discrepancy between re-substitution and validation performance suggests model overfitting. The potential for false signals is high if problems such as data leakage exists.

Although time-to-event method has better performance than binary classification approach, the improvement in the holdout validation set is small. An entirely separate test set has been withheld and will be used to test our final models at conclusion of the INCISE project.

References

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